

■ Learning Path — FastAPI + LangGraph + Supabase

This guide curates essential resources to master the backend stack you're using: Firebase Gateway + LangChain Orchestrator (FastAPI + LangGraph) + Supabase (pgvector).

Each section includes recommended documentation, tutorials, and videos for building your MVP quickly and solidly.

1■■ FastAPI — Modern Python Web Framework

****Official Docs:**** <https://fastapi.tiangolo.com/>

****Korean Tutorial:**** <https://wikidocs.net/book/8531>

****Videos:****

- 1 • Fireship — Build and Deploy FastAPI to Google Cloud Run (12 min)
- 2 • freeCodeCamp — FastAPI Crash Course (2 hr)

****Core Concepts:**** Pydantic models, async/await, dependency injection, middleware, background tasks.

2■■ LangGraph (LangChain) — Agentic Workflow Framework

****Docs:**** <https://langchain-ai.github.io/langgraph/>

****LangChain Basics:**** <https://python.langchain.com/docs/tutorials>

****Videos:****

- 1 • LangGraph Tutorial by Harrison Chase
- 2 • LangChain Crash Course (Greg Kamradt / Data Independent)
- 3 • LangGraph Concepts (LangChain Devs, 2024)

****Key Idea:**** Define graph nodes as stateful steps (plan→search→evaluate→synthesize). LangGraph handles transitions, retries, and checkpoints.

3■■ Supabase — Postgres + pgvector + Auth/Storage

****Docs:**** <https://supabase.com/docs>

****Korean Guide:**** Nomad Coders Supabase for Beginners

****Videos:****

- 1 • Fireship — Supabase Crash Course (10 min)
- 2 • freeCodeCamp — Supabase Full Course (2 hr)
- 3 • How to use pgvector with Supabase (LangChain Integration)

****Core Concepts:**** Postgres with pgvector for semantic search, RLS for user data isolation, integrated file storage API.

4■■ Combined Learning Roadmap (2 Weeks Sprint Plan)

- 1 **Week 1:** FastAPI + Supabase basics, build /ingest/url endpoint and DB connection.
- 2 **Week 2:** LangGraph workflow creation (planner → searcher → summarizer). Integrate Firebase call chain.
- 3 **Week 3:** Connect Android app (Retrofit) to Firebase Gateway + Orchestrator.
- 4 **Week 4:** Extend to RAG + Mind Map visualization.

5■■ Advanced Resources

- 1 • Cloud Run Deployment: <https://cloud.google.com/run/docs/tutorials/fastapi>
- 2 • LangGraph Cookbook: <https://langchain-ai.github.io/langgraph/examples/>
- 3 • LangChain + Supabase Integration:
<https://python.langchain.com/docs/integrations/vectorstores/supabase>

■ Most Similar Demo Example

LangChain Blog: Build a Personalized RAG Agent with Supabase (2024)

■ <https://blog.langchain.dev/build-a-personalized-rag-agent-with-supabase/>

This example closely matches your architecture (Firebase → Orchestrator → Supabase with pgvector) and demonstrates personalized retrieval, LangGraph orchestration, and Supabase storage integration.

■ Tip: Follow this progression—start with FastAPI endpoints locally, connect to Supabase using service key, and then integrate LangGraph workflows. Once stable, deploy Orchestrator on Cloud Run and connect Firebase Functions as a gateway.